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TRIAL 3: CF TOP PLATE DROP

● PASS VERIFIED

Does the 8-ply 3K twill top plate crack or flex into the blade when the bot lands flat from 1 m?

PLATE SURVIVES A 1 M FLAT LANDING? PASS

YES

first-ply SF 4.08, flex 2.33 mm

PLATE CRACKS PASS

SF 4.08

P(fail) 0% | SF P5 2.74

PLATE HITS BLADE [GOVERNING] PASS

SF 2.14

P(fail) 0% | SF P5 1.44

Margin to each user-defined failure mode



TRIAL 3: CF TOP PLATE DROP

Time histories of the governing states (nominal case)

Plate force from the motor

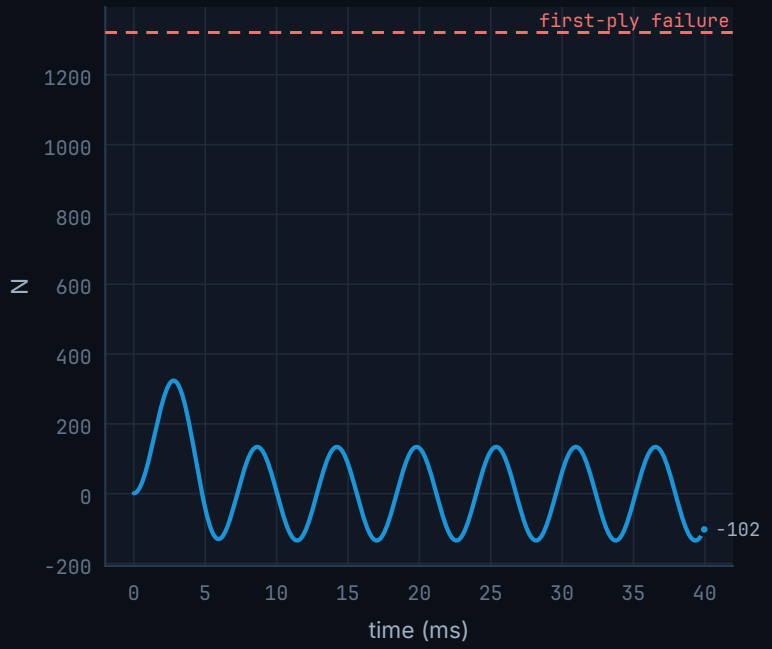
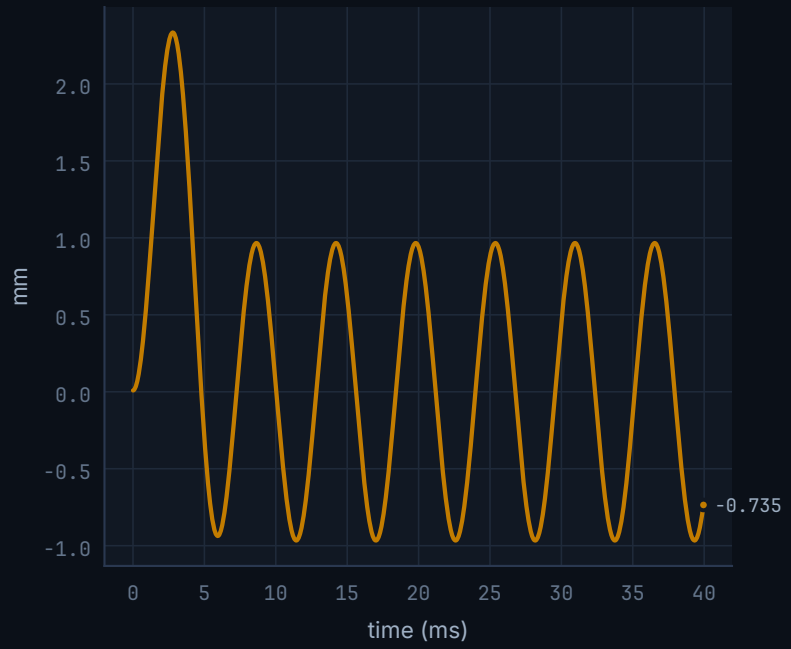
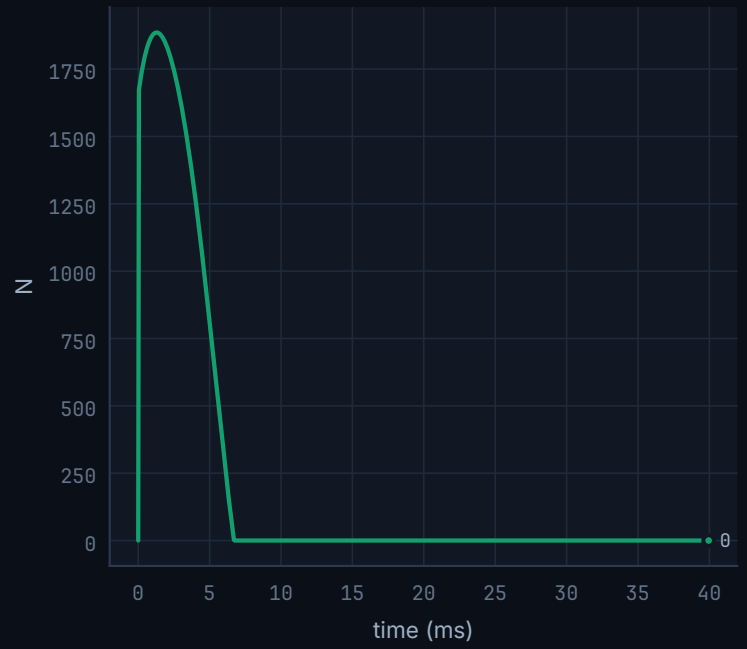


Plate flex under the motor



Floor force on the wheels



TRIAL 3: CF TOP PLATE DROP

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'plate hits blade' margin



Plate crack margin vs drop height



SIMBLOCKS // VERIFICATION

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TRIAL 3: CF TOP PLATE DROP

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	2 bounds respected
stability_margin	● PASS	critical dt 0.00186 s / dt 2e-06 s = 929.8x
dt_convergence	● PASS	F_plate_peak_N: err 0.043% (order 1); plate_defl_mm: err 0.043% (order 1); F_floor_peak_N: err 0.044% (order 1)
energy_ledger	● PASS	worst closure residual 0.072% across 3 dt levels (tol 5%)
limit:no gravity, no drop → no load	● PASS	matches expected limit
knockout:rigid plate	● PASS	F_plate_peak_N -3.19%, plate_defl_mm -100%, F_floor_peak_N +2.59%
knockout:no floor damping	● PASS	F_plate_peak_N -5.52%, plate_defl_mm -5.52%, F_floor_peak_N +21.7%
sensitivity	● PASS	4 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	200 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● PASS	plate cracks: nominal SF 4.08, P(fail) 0%; plate hits blade: nominal SF 2.14, P(fail) 0%